Smashing Memory & I/O Limits: Architectures for the GenAI Era



OCP GLOBAL SUMMIT

October 13-16, 2025
San Jose, California

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Smashing Memory & I/O Limits: Architectures for the GenAI Era

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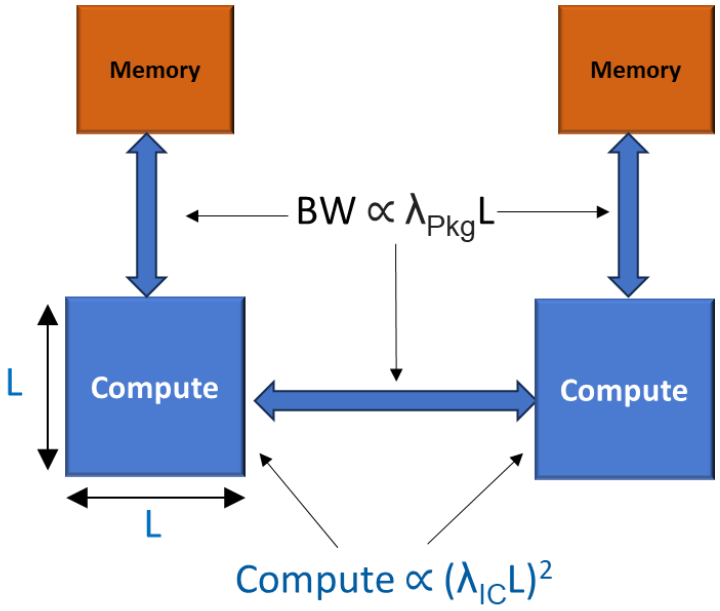
eliyan

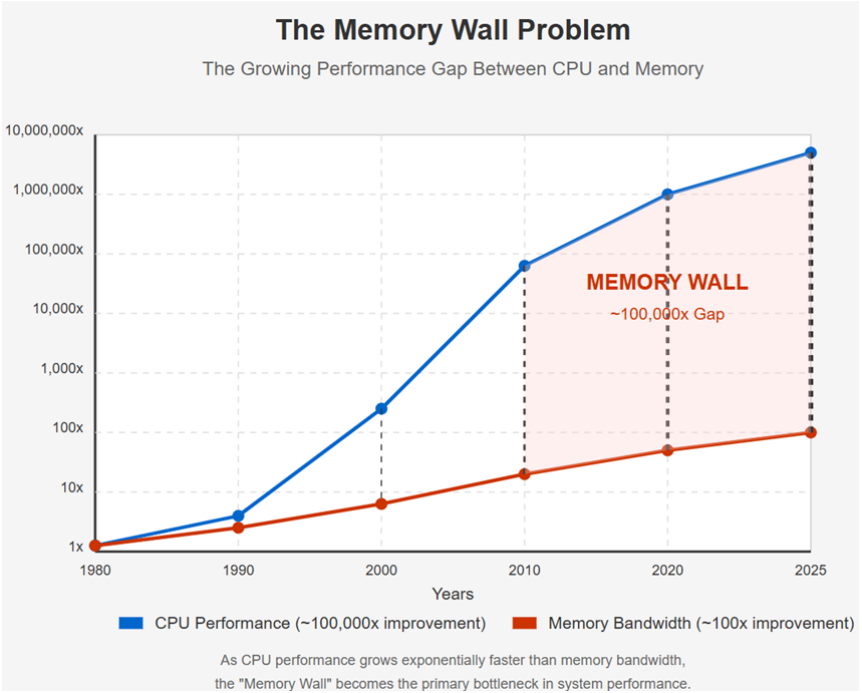
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Presentation Outline

- Performance Walls
  - Power Wall, Bandwidth Walls: Memory & I/O Walls
- SBD: A novel D2D technology for superior Bandwidth/Power
  - Implementation, silicon results, comparison
- High-performance D2D interconnects: The solution to Performance Walls
  - Advantages of 2.0D D2D: Long reach connectivity on large Std. substrates
  - Multi-port Custom HBM (cHBM) to boost Memory & I/O bandwidths
- Performance limitations at system level, beyond the package
  - Leveraging long reach 2.0D D2D to enable ultra high-BW AI or Switch ASICs
  - GPU ASIC with 32TB/s memory bandwidth & 50T-100Tbps I/O bandwidth
  - Switch ASIC with 400Tbps of aggregate bandwidth
    - Leveraging SBD technology to double electrical port bandwidth: 224G → 448G
- Call to Action

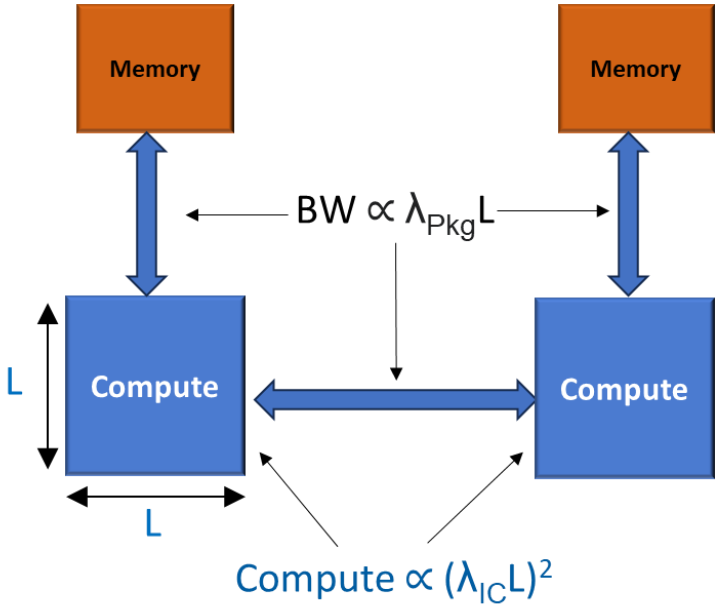
Top Bottlenecks in AI Performance: Memory Wall

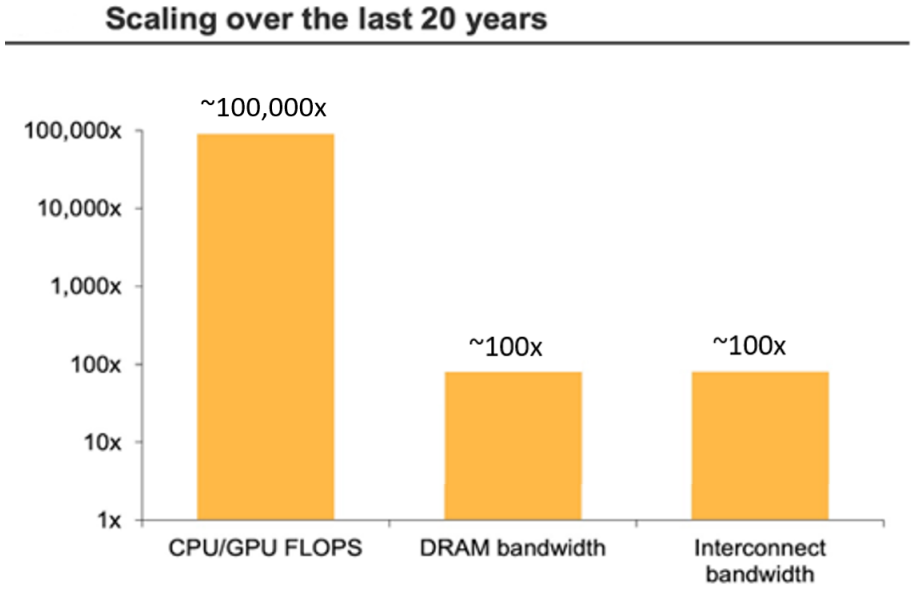




Disparity in **bandwidth gains** vs **compute gains** is the top bottleneck for performance

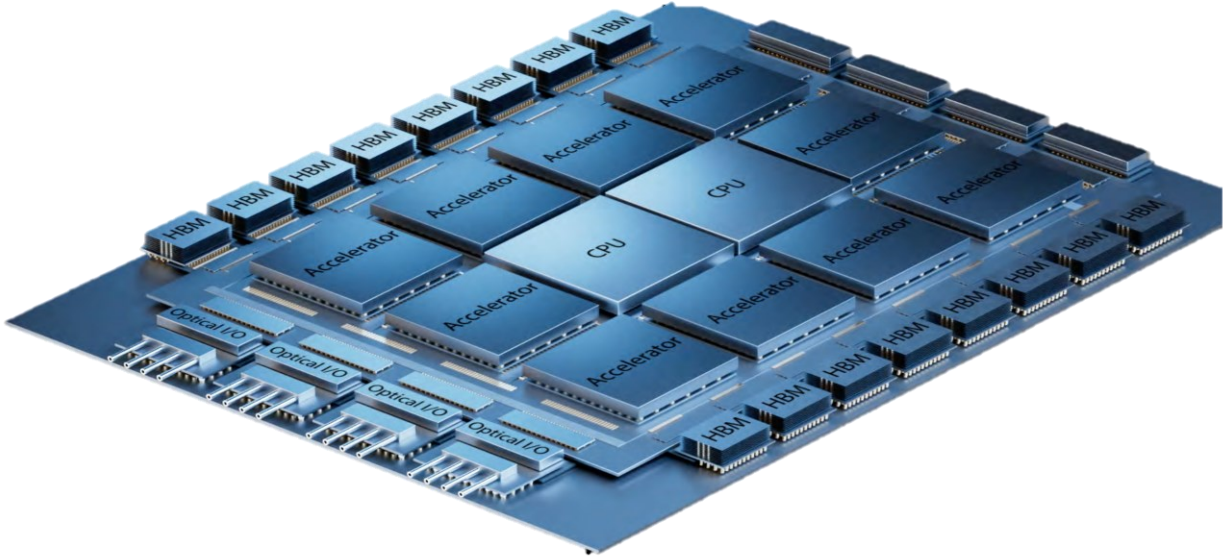
Top Bottlenecks in AI Performance: Memory Wall & I/O Wall





Disparity in **bandwidth gains** vs **compute gains** is the top bottleneck for performance

Move to Chiplet-based System to Maximize Bandwidth/Power



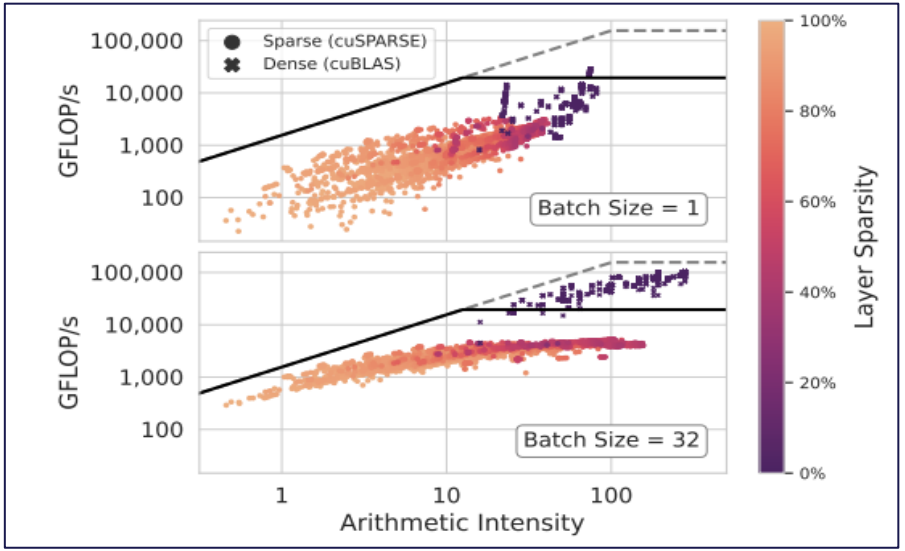
- Ultra high-performance AI systems need:
  - 10s of Tbps interconnect bandwidth between processor & memory devices
  - The power for these interconnects must be ultra low to be practical
- Chiplet interconnects on a package substrate offer the best bandwidth/power ratio: >1Tbps/W
- AI systems with superior BW/power need very large pkg substrate with as many chiplets as possible

Gen AI Hardware Still Facing the Memory Bandwidth Wall

A Memory BW Wall Example → Blackwell 200

- B200 Compute Performance = 10,000 TFLOPS for FP8.
- B200 Max Memory Bandwidth = 8TB/s
  - ➔Up to 1250 FLOPs per every Byte from Memory
  - ➔Even at very high arithmetic intensity of 500*, B200 will be utilized at ~50% (=500/1250).

* The arithmetic intensities (FLOPS/Bytes) of the AI models should fall in the range 1-500.



Roofline model for Sparse vs. Dense:
Measuring throughput of SpMM on unstructured sparse layers and GEMM on dense layers from all trained models in A100.

Solid line: CUDA core peak throughput.

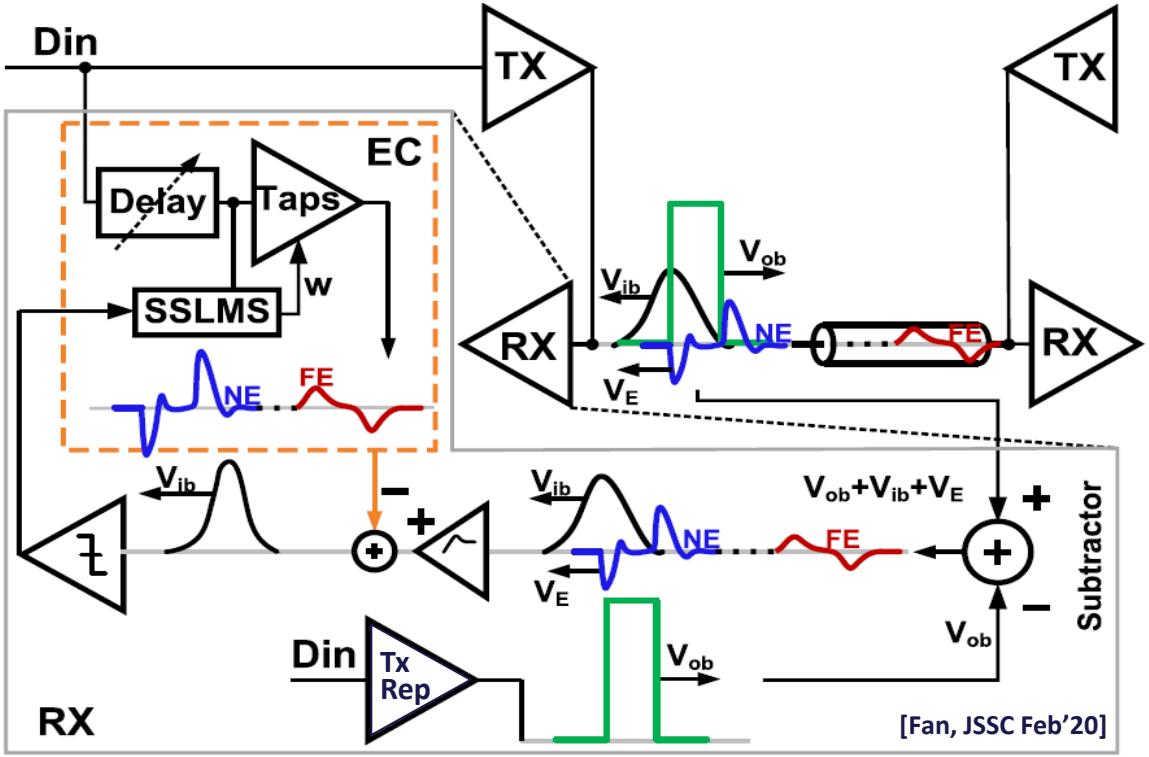
Dashed line: Tensor core peak throughput

Source: "The Sparcity Roofline: Understanding the Hardware Limits of Sparse Neural Networks," by Cameron Shinn, et al.

- Today's chiplet interconnects still NOT sufficient to provide enough memory/ I/O bandwidth to AI XPUs
- Higher performance chiplet interconnects and innovative architectures still required

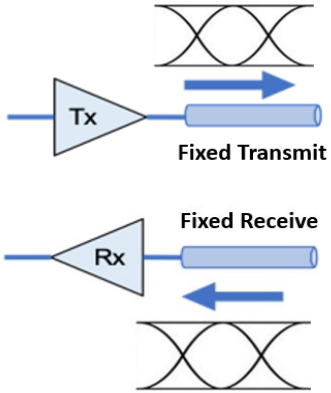
Simultaneous Bi-Directional (SBD) Signaling

- V_{ob}: Local Transmitted Signal
- V_{ib}: Far-end Received Signal
- V_E: Echo Signal (VNE + VFE)
- NE: Near-end Echo
  - From local discontinuities (Vias, etc.)
- FE: Far-end Echo
  - From remote discontinuities (term mismatch, etc.)
- EC: Echo Canceller (Multi-tap FIR)



Die-to-Die (D2D) Interconnect Signaling Choices and Tradeoffs

UD[1] NRZ
N Gbps @ N Gbaud



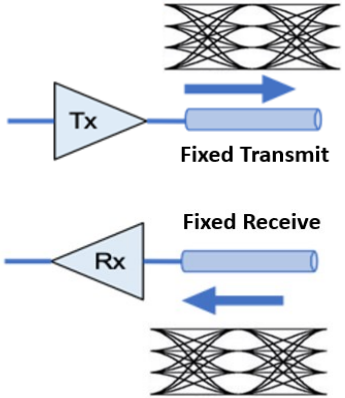
Cons

- Low bandwidth per lane: 1 bit/baud
  - ➔ data rate = baud rate
- Higher baud rates ➔ Complex Tx/Rx
  - ➔ Higher power

Pros

- High-SNR NRZ ➔ Simpler Tx/Rx design
  - ➔ Lower power

UD PAM4
2xN Gbps @ N Gbaud



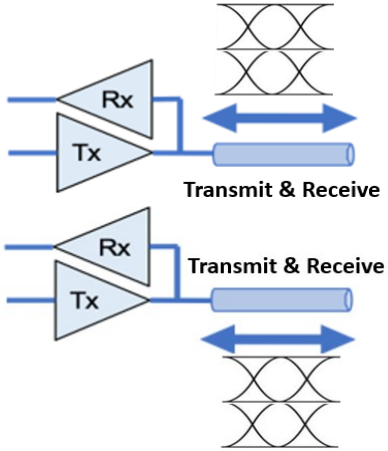
Cons

- Low-SNR PAM4 ➔ Complex Tx/Rx + FEC
  - ➔ Higher power & latency

Pros

- Higher bandwidth per lane: 2 bits/baud
  - ➔ data rate = 2x baud rate

SBD[2] NRZ
2xN Gbps @ N Gbaud



Cons

- Echo cancellation complexity

Pros

- Higher bandwidth per lane: 2 bits/baud
  - ➔ data rate = 2x baud rate
- High-SNR NRZ ➔ Simpler Tx/Rx design
  - ➔ Lower power

1- UD: Uni-Directional, **2**- SBD: Simultaneous Bi-Directional

Die-to-Die (D2D) Interconnect Signaling Comparison

	UD NRZ	SBD NRZ (A)	UD PAM4	SBD NRZ (B)
Signaling baud rate	R	R/2	R	
Data rate/wire	R	R	2 x R	2 x R
Number of D2D lanes	N	N	N	N
Error Correction (BER<1E-12)	None	None	YES	None
Total Throughput (over N Lanes)	**N x R**	**N x R**	**2 x N x R**	**2 x N x R**
Signaling SNR[1]	Good	Good	Low	Good
Performance/Cost Tradeoffs	Reference	Same throughput Lowest power[2]	Double throughput Highest power & latency	Double throughput Similar power[3]

1- High signaling SNR enables connectivity over higher loss → longer reach.
2- Half baud rate enables PHY circuits to run at half clock rate, thus considerably lower Vdd → Leads to Vdd^2 power saving.
3- SBD burns slightly (<10%) more power for Echo cancellation.

NuLink: Eliyan's D2D Chiplet Interconnect Technology

Three generations of silicon-proven D2D PHYs over Std Pkg (SP) deliver long reach connectivity (5mm-50mm)

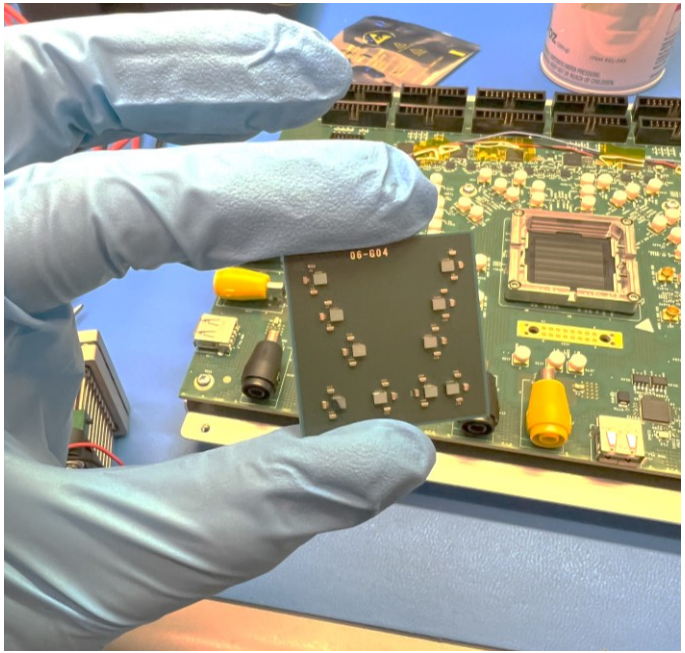
NuLink-SP0: 28Gbps (SBD: 2x14Gbps)
- Two generations of test chips for the original design
- 2020: Production. Shipped >10M to date

NuLink-SP1: 40Gbps (SBD: 2x20Gbps)
- 2023: Silicon proven in 5nm

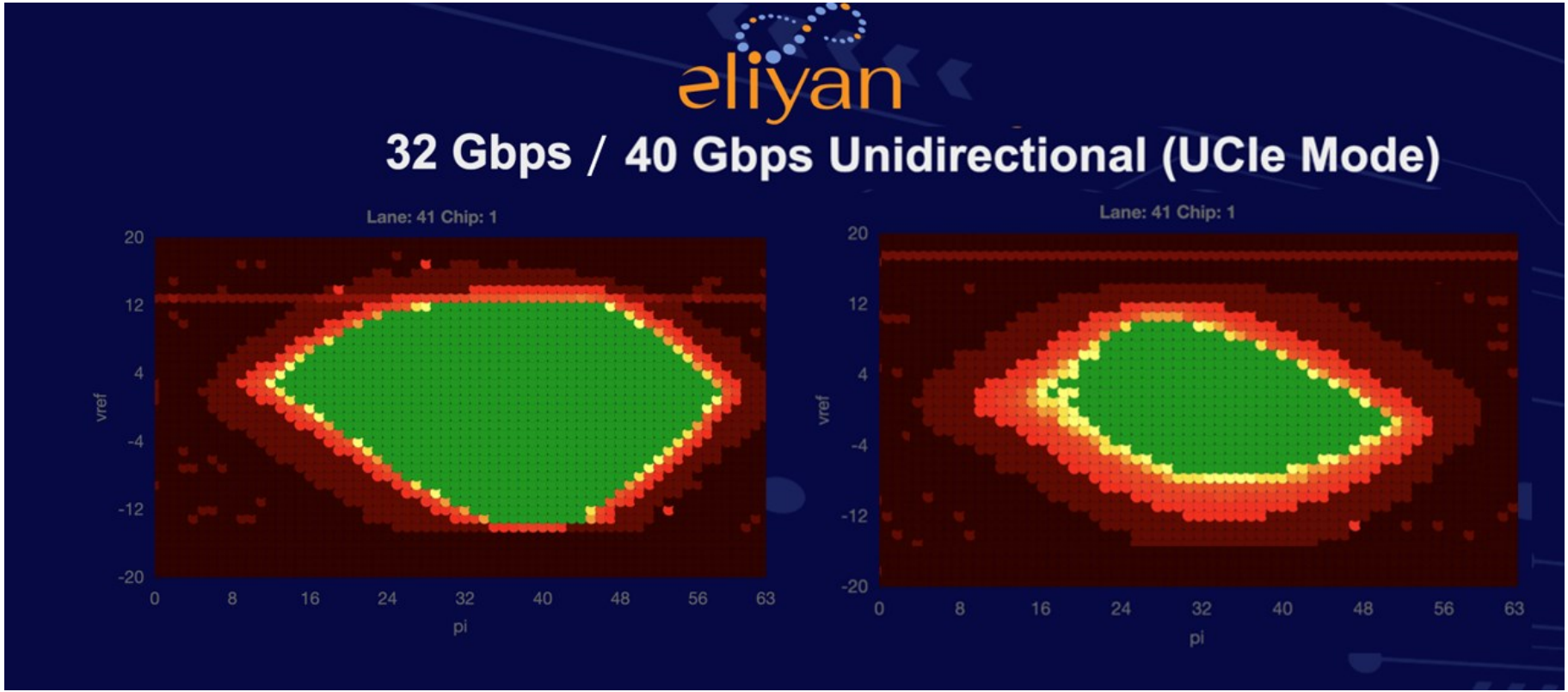
NuLink-SP2: 64Gbps (SBD: 2x32Gbps)
- 2024: Silicon proven in 3nm

NuLink-AP2: 64Gbps (SBD: 2x32Gbps)
- 2026: Target tapeout

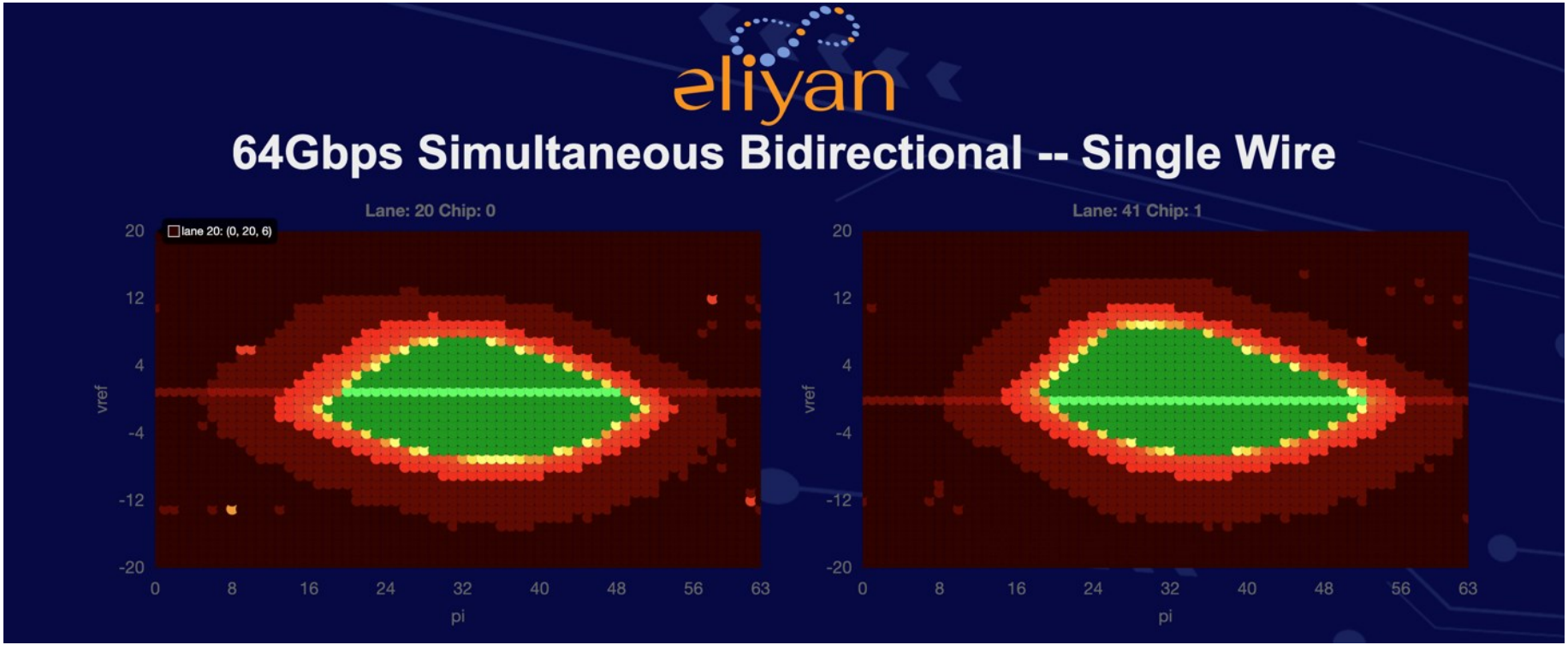
NuLink-SP3: 100Gbps
- 2026: Target tapeout



NuLink-SP2: 32Gbps/40Gbps UD-SP Measured Eye Diagrams over SP Substrate



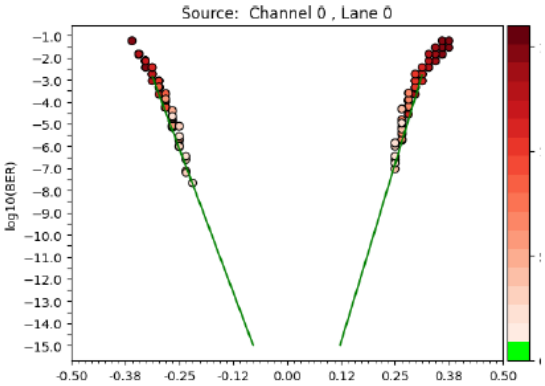
NuLink-SP2: 64Gbps SBD-SP Measured Eye Diagram over SP Substrate



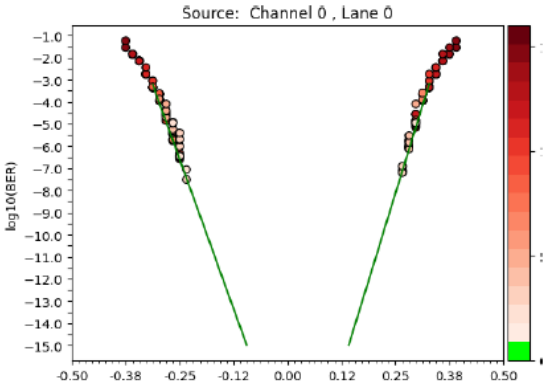
NuLink-SP2: 64G SBD-SP Bathtub Curves and BER

Slow Corner 3nm Silicon Measurements across 0C/50C/100C

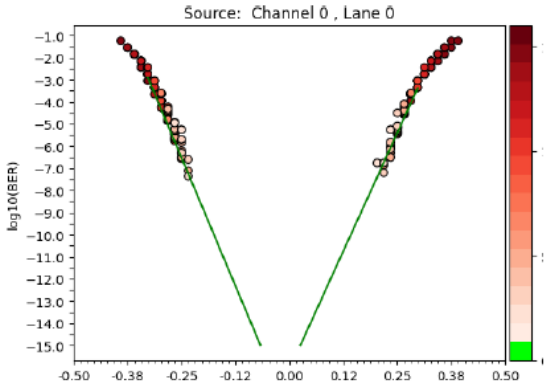
Cold



Nom



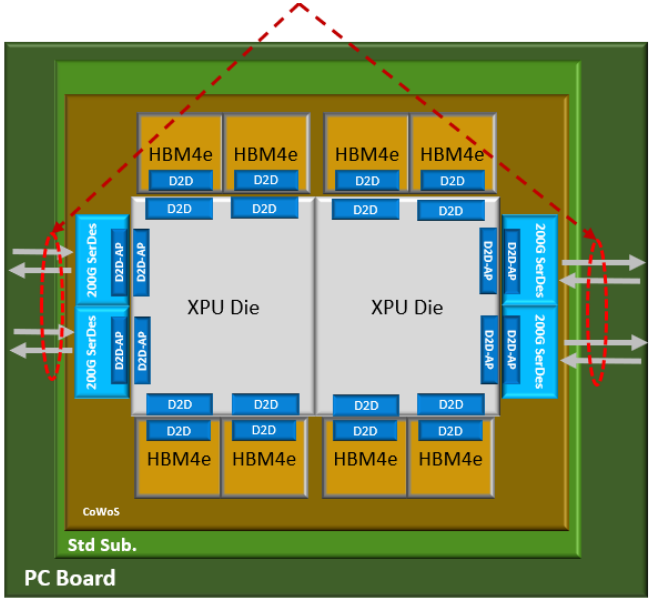
Hot



- Link initialized at nominal temperature, then performs temp sweep
- 64G SBD native PHY (without FEC) delivers BER < 10^{-15}

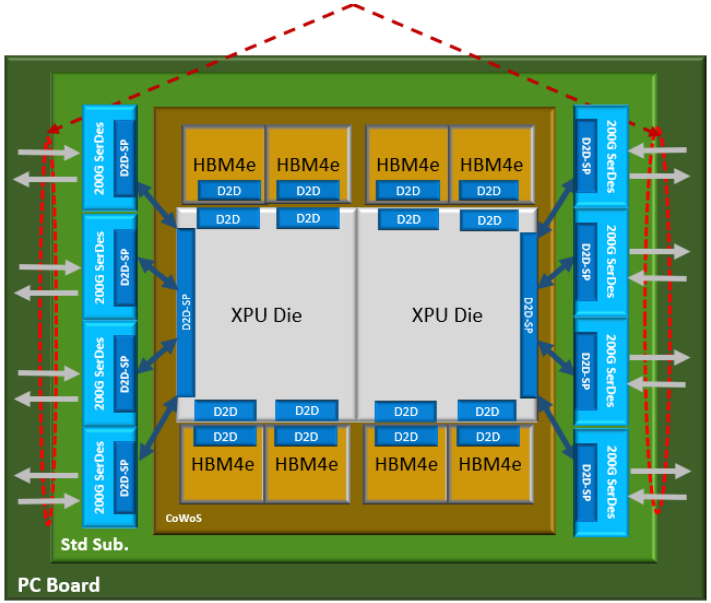
Example 1: Leveraging 64G D2D-SP Long-Reach (LR) to Quadruple XPU I/O Bandwidth

Aggregate IO BW → 24Tbps-32Tbps



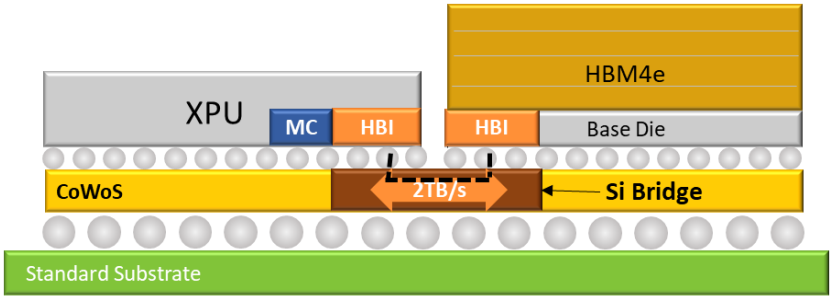
64Gbps D2D-SP

Aggregate IO BW → 96Tbps-124Tbps

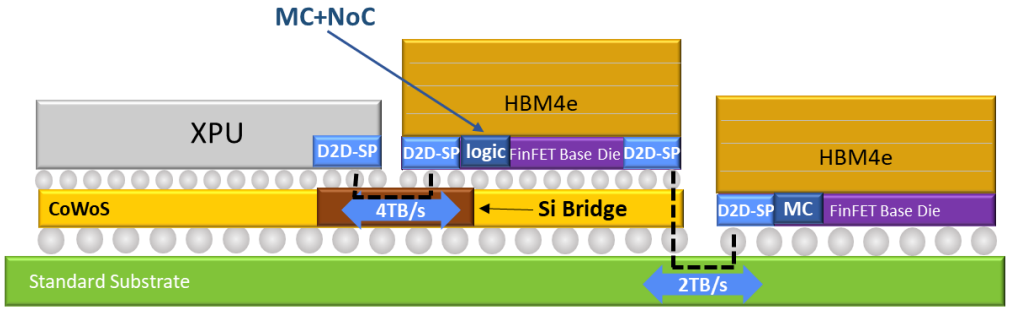


- D2D PHYs on Adv Pkg provides high BW density but limits the peripheral chiplets connected to an XPU
- 64G→ 100Gbps D2D-SP provides high beachfront BW density over long reach (5mm-50mm), enabling fan-out topologies that can almost quadruple (~4x) the peripheral chiplets connected to an XPU

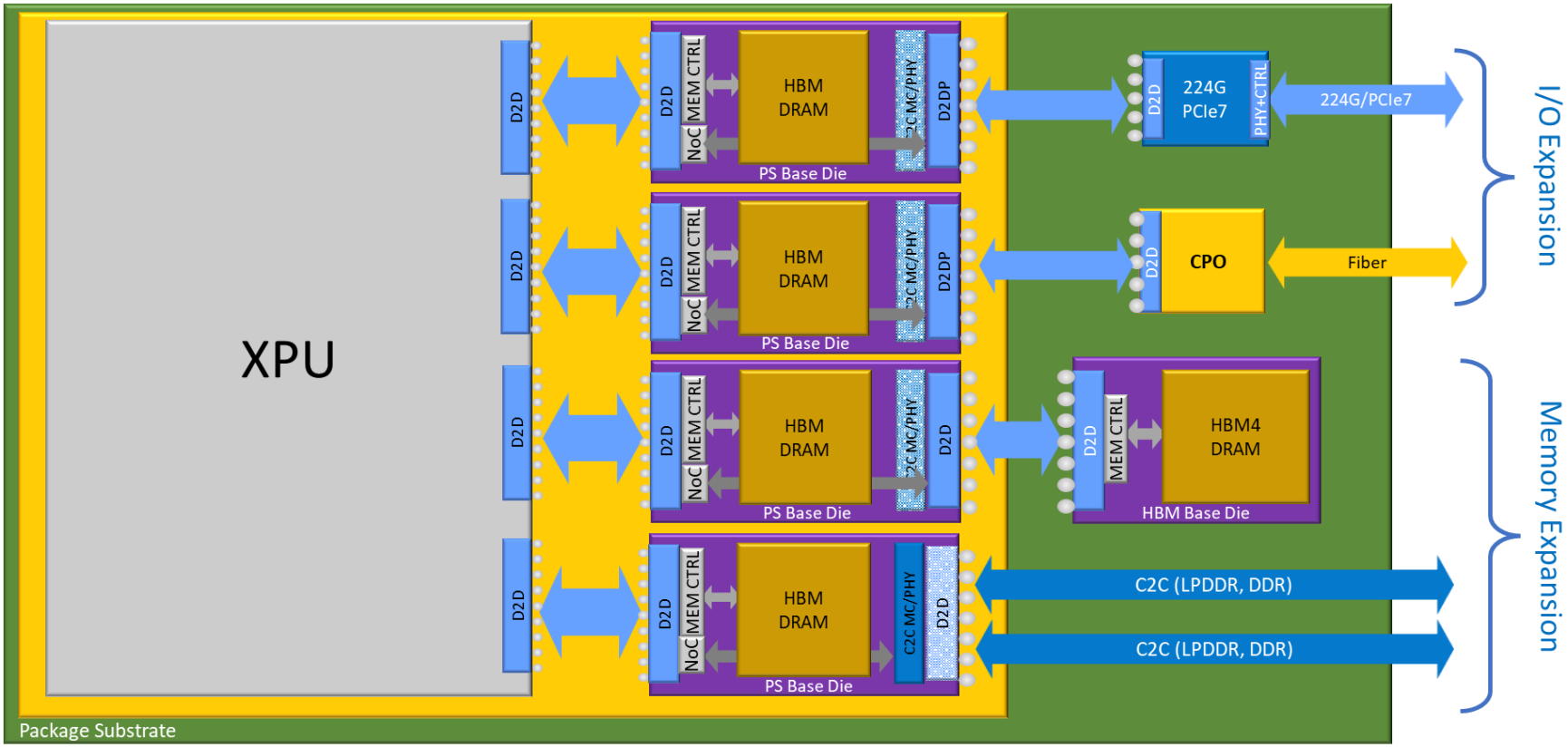
Custom HBM Offers Multi-Port Base Die → Memory & I/O Expansion



- A 64G SBD provides enough bandwidth to support 2xHBM4es within a HBM beachfront (11mm)
- Multi-port cHBM base die with SBD PHYs enables memory expansion by daisy chaining of HBMs to break the Memory Wall
- SBD-SP at <0.5pJ/bit can support HBM4e bandwidth on Std Pkg while meeting HBM thermal requirement
  - ➔ No need for large CoWoS to expand memory

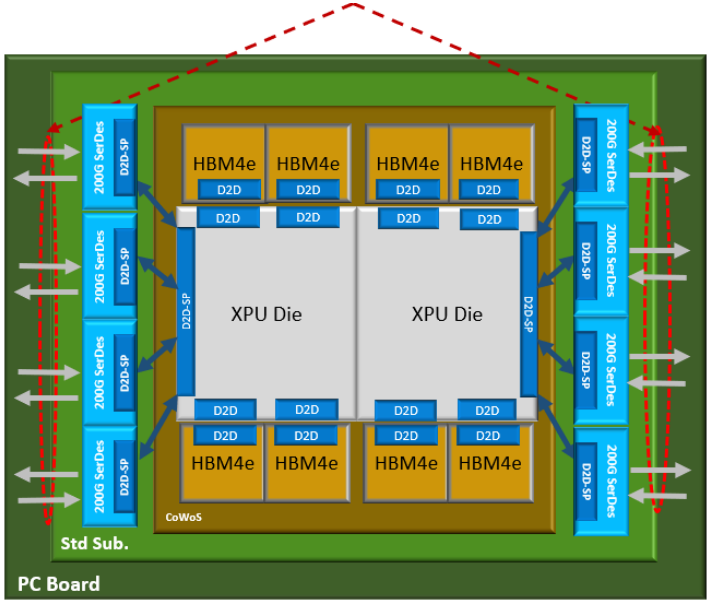


Multi-port cHBM Empower Higher Performance AI Architectures



Example 2: Leveraging Multi-port cHBM + 64G D2D-SP to Further Double XPU I/O Bandwidth

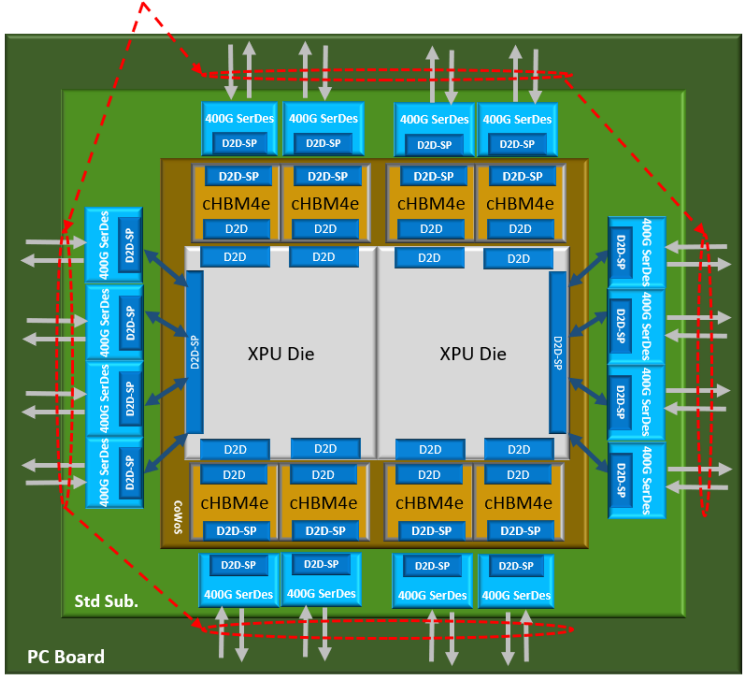
Aggregate IO BW → 96Tbps-124Tbps



64Gbps D2D-SP

Multi-Port cHBM

Aggregate IO BW → 192Tbps-248Tbps

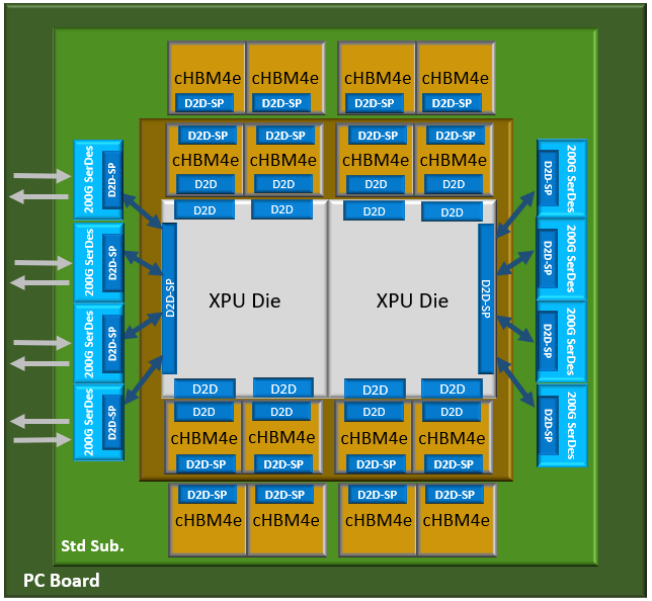


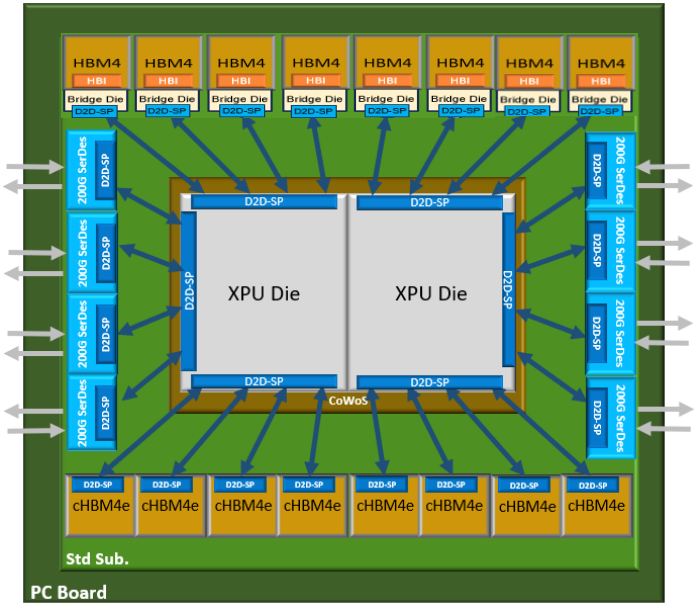
- Multi-port cHBM enables all XPU beachfronts to be used for off-chip connectivity, thus providing another factor of ~2X I/O bandwidth

→ Eliminates the need for CPO in the next generation AI systems to increase XPU bandwidth!

Example 3: Leveraging 64G D2D-SP Combined with HBM to Achieve 2x HBM & 4x I/O Bandwidth

Aggregate IO BW: 24T-32Tbps → ~96T-124Tbps + HBM BW: 16TB/s (8 HBM4e) → 32TB/s (16 HBM4e)

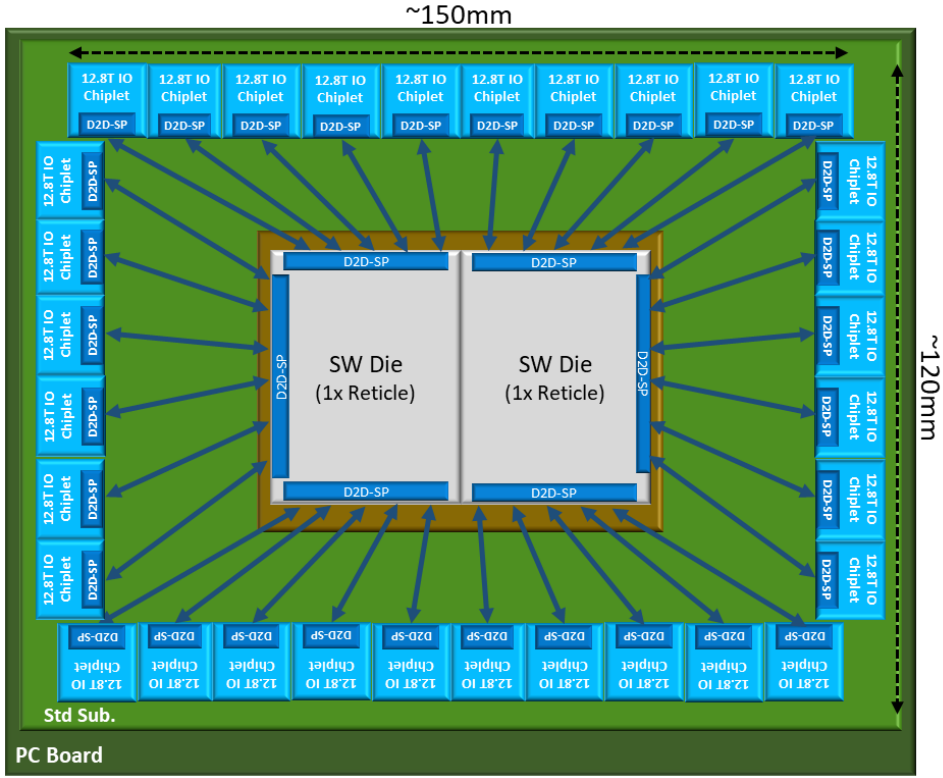




- 64Gbps D2D-SP between HBMs and XPUs enables doubling HBM chiplets per XPU, either by daisy changing them or by fanning out to 2x HBMs over a wider beachfront.
- 64Gbps D2D-SP can be implemented in the Base die of cHBM or in a separate Bridge Die that converts standard JEDEC HBM interface to 64Gbps D2D-SP

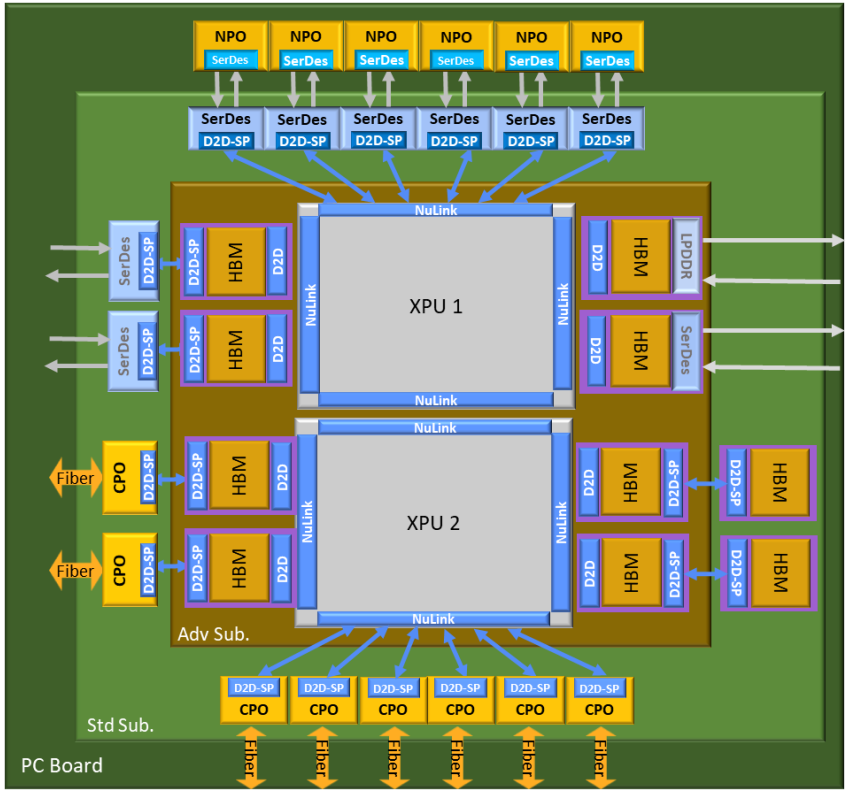
Example 4: Leveraging Next-Gen 100Gbps D2D-SP LR to Build 400Tbps Switch ASIC

- NuLink-SP (64Gbps→ 100Gbps) delivers of 4T-6Tbps/mm combined (Tx+Rx) bandwidth density, or 2T-3Tbps/mm per direction on standard substrates bump pitch=110um
- Two reticle sized Switch ASICs can achieve a total full-duplex bandwidth of >200Tbps with NuLink-SP at 80Gbps with bandwidth density of 2.5Tbps/mm/direction
- NuLink-SP long reach enables fanning out to 32x 12.8Tbps IO chiplets, achieving full-duplex bandwidth of 409.6Tbps.
- The 12.8T IO chiplets can be
  - CPO @200Gbps/lambda
  - SerDes @ 400Gbps/lane (400G PAM4 SBD, 400G PAM6)



Call to Action

- Long-reach 64G-100Gbps D2D on SP substrate provides significant BW densities that can eliminate the performance bottlenecks of future AI ASICs
- 64G-100Gbps D2D-SP with long reach (LR) together with cHBMs maximize XPUs total bandwidth efficiency, thus:
  - Breaks the Memory Wall
    - More HBMs per XPU with smaller CoWoS delivering higher memory bandwidth & capacity
  - Breaks the I/O Wall
    - I/O chiplet connectivity on 4 XPU edges, even edges with connected HBMs
    - D2D-SP LR enables fanning out to large number of I/O chiplets, quadrupling the IO bandwidth
- Let's work together to smash the Memory & I/O Walls and enable creation of super large AI ASICs



*Solutions presented are protected by Eliyan's patent portfolio

Thank You!

